

Interpolation

Recap

Lagrange interpolation

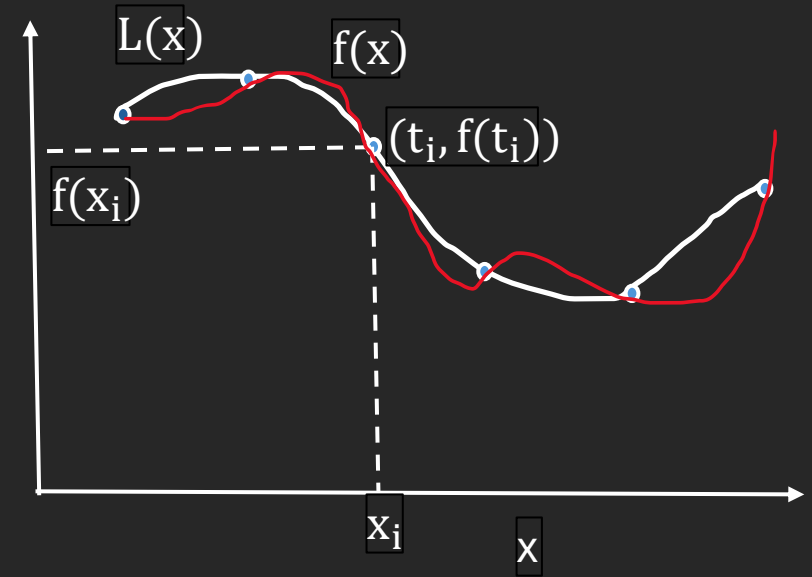
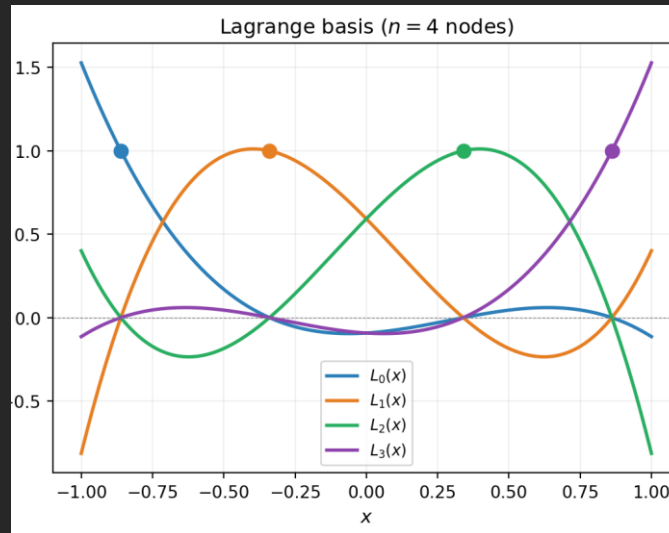
Given a set of coordinate pairs $(x_i, f(x_i))$, $f(x)$ can be interpolated by a function $L(x)$, such that

$$L(x_i) = f(x_i)$$

$$L(x) = \sum_{i=1}^{k+1} f(x_i) l_i(x)$$

$$l_i(x) = \prod_{m=1, m \neq i}^{k+1} \frac{(x - x_m)}{(x_i - x_m)}$$

$$l_i(x_j) = \delta_{ij}$$

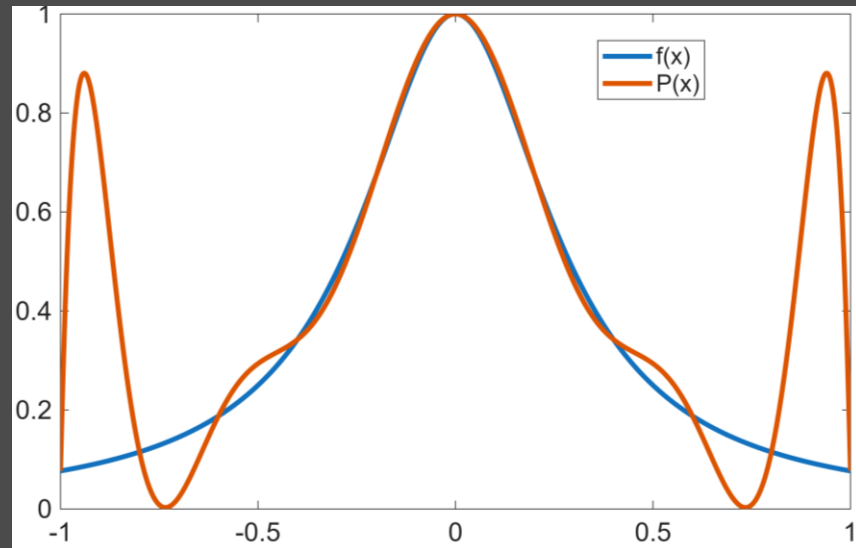


Newton interpolation

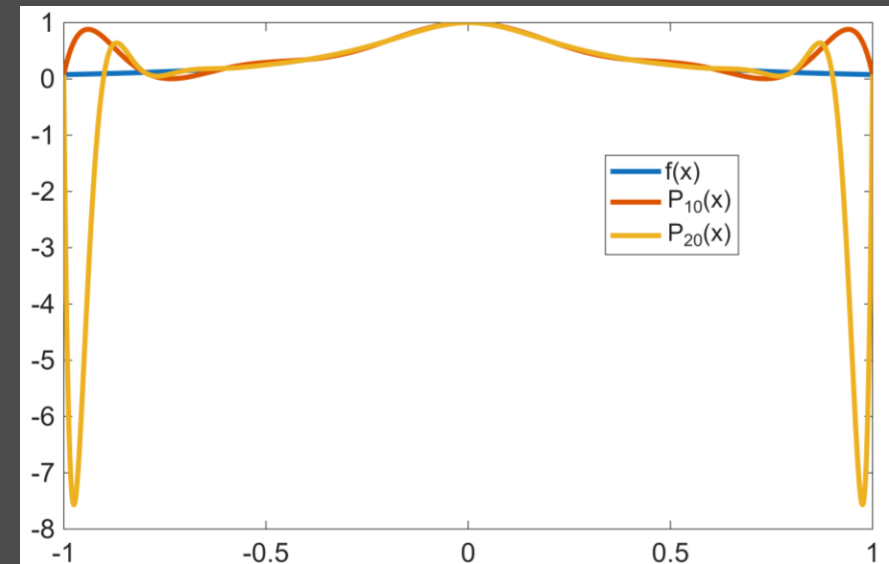
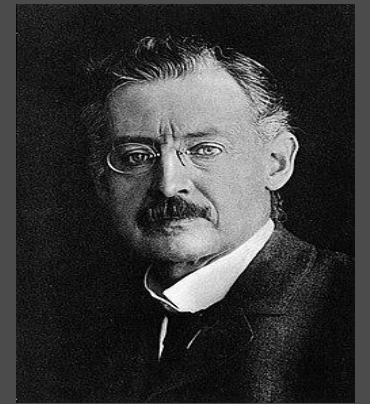
$$\begin{aligned} P(x) = & f[x_1] + f[x_1 \ x_2](x - x_1) \\ & + f[x_1 \ x_2 \ x_3](x - x_1)(x - x_2) \\ & + f[x_1 \ x_2 \ x_3 \ x_4](x - x_1)(x - x_2)(x - x_3) \\ & + \dots \\ & + f[x_1 \ \dots \ x_n](x - x_1) \dots (x - x_{n-1}). \end{aligned}$$

Runge Phenomenon

$$f(x) = \frac{1}{1 + 12x^2}$$

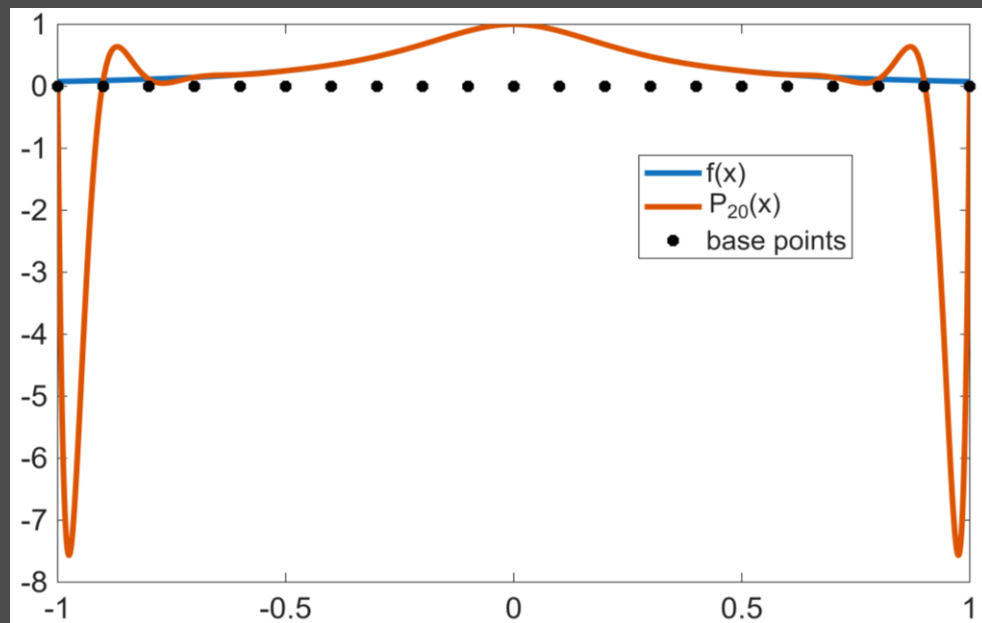


Carl Runge
(1896-1927)



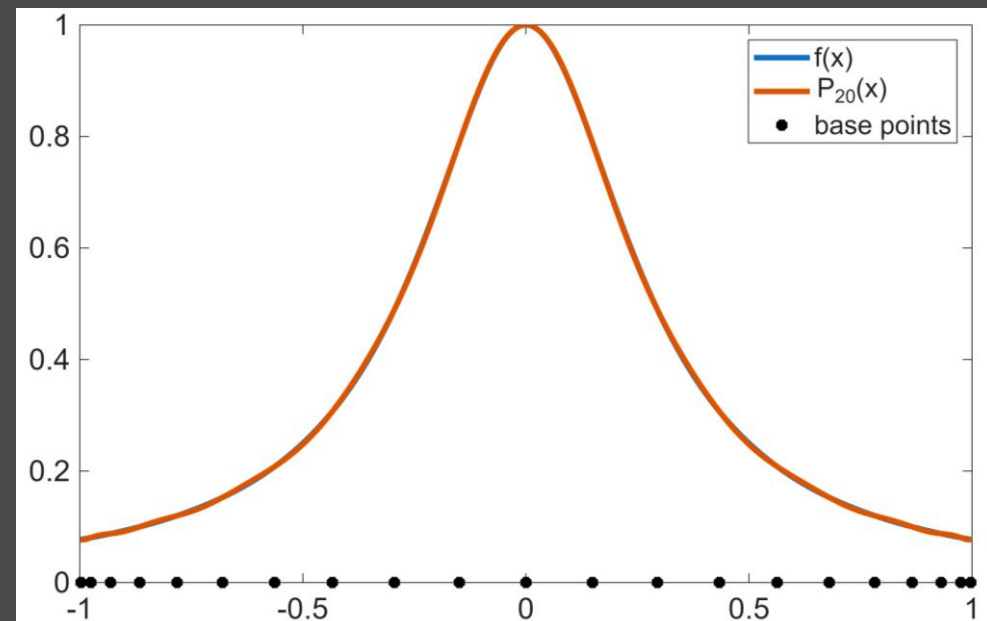
Unevenly spaced nodes/data points improve interpolation

$$f(x) = \frac{1}{1 + 12x^2}$$



Equispaced base points

(x values of the data points)



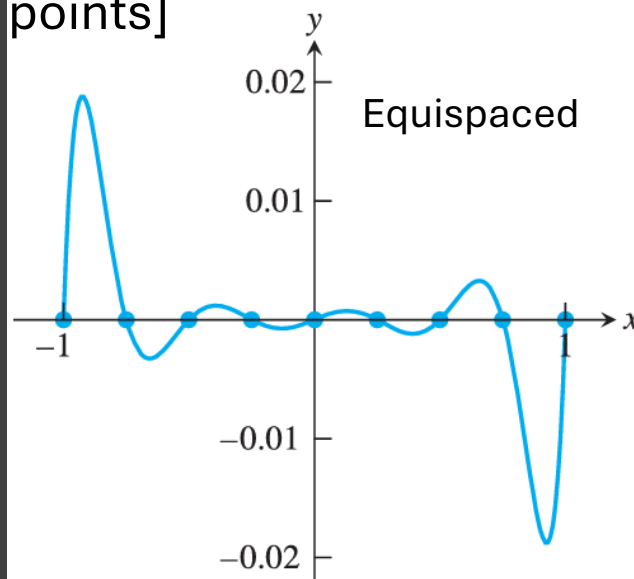
Nonequispaced base points

Reason behind Runge phenomenon:

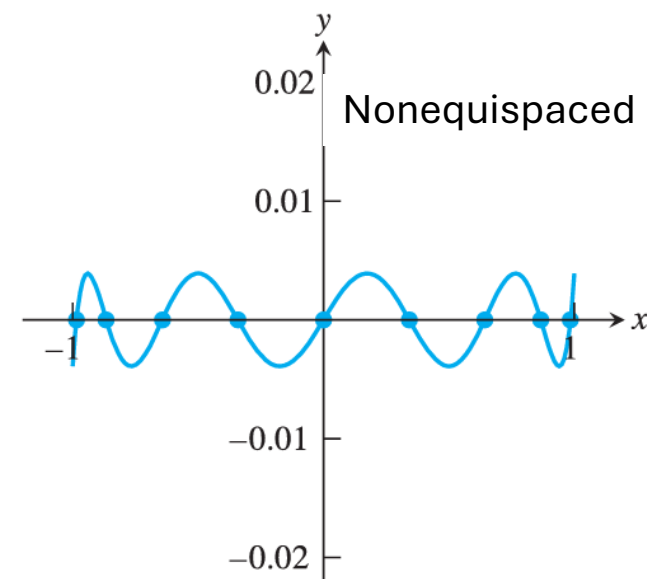
Interpolation error =

$$\frac{(x - x_1)(x - x_2) \cdots (x - x_n)}{n!} f^{(n)}(c)$$

$y = (x - x_1)(x - x_2) \cdots (x - x_9)$ [9 base points]



(a)



(b)

Chebyshev theorem:

In the interval $[-1, 1]$, $|(x - x_1)(x - x_2) \dots (x - x_n)|$ is minimized if

$$(x - x_1)(x - x_2) \dots (x - x_n) = \frac{T_n(x)}{2^{n-1}},$$

$$\longrightarrow x_i = \cos\left(\frac{(2i - 1)\pi}{2n}\right), \text{ roots of } T_n(x)$$

$T_n(x)$ = Chebyshev polynomial
 n = degree of the polynomial

T_n 's are solutions of
Sturm – Liouville equation:

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y = -\lambda w(x)y$$

Properties of Chebyshev function T_n :

(1) T_n 's are polynomials:

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

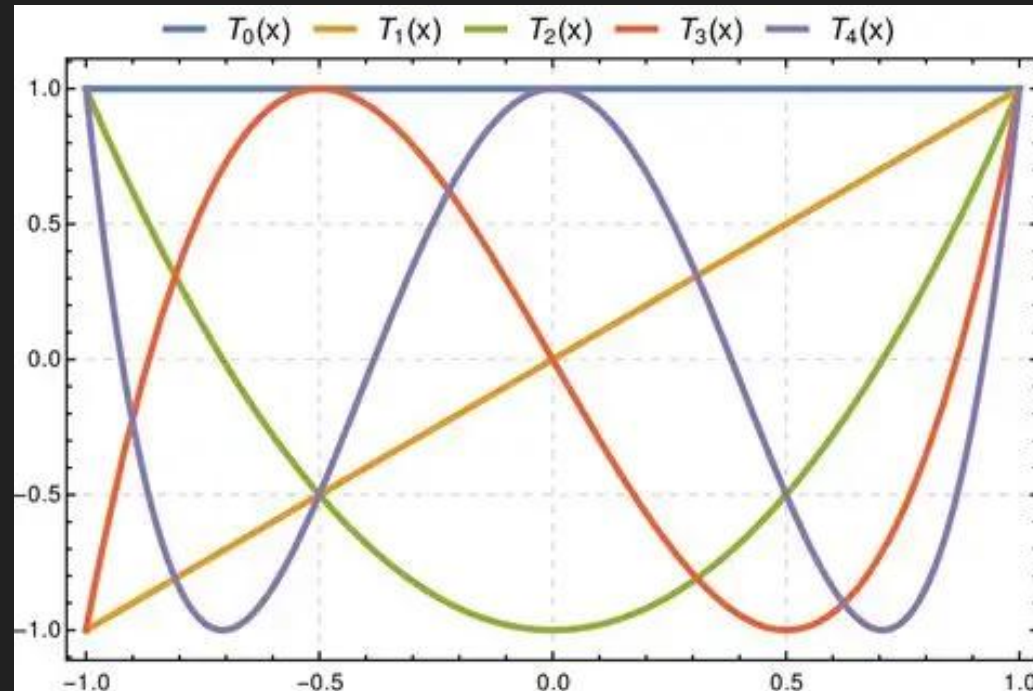
$$T_3(x) = 4x^3 - 3x.$$

$$T_n(x) = \cos(n \cos^{-1}(x))$$

(2) $T_n(x)$ is an n – degree polynomial and the leading coefficient is 2^{n-1}

(3) The max. value of $T_n(x)$ is 1, i. e., $|T_n|=1$

(4) $T_n(x)$ alternates between 1 and -1 $n + 1$ times in the interval $[-1, 1]$



Chebyshev's theorem: If $(x - x_1)(x - x_2) \dots (x - x_n) = \frac{T_n(x)}{2^{n-1}}$, then $|(x - x_1)(x - x_2) \dots (x - x_n)| = \frac{1}{2^{n-1}}$

Proof: Assume there is a **monic** polynomial $P_n(x)$, such that $|P_n(x)| < \frac{1}{2^{n-1}}$.

Monic polynomials are polynomials where the coefficient of the highest – degree monomial is one.

Example: $x^2 + bx + c$ (monic) ; $2x^2 + bx + c$ (not monic)

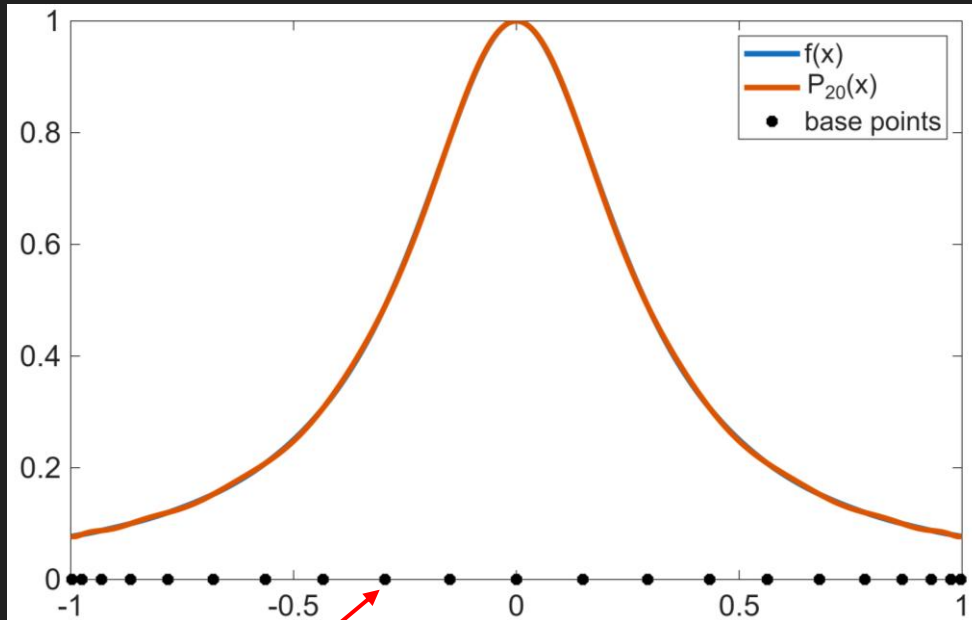
Since T_n alternates betwn. -1 and 1 $n + 1$ times, at these points $P_n - T_n/2^{n-1}$ alternates betwn. $-ve$ and $+ve$.

This implies that $P_n(x) - T_n(x)/2^{n-1}$ crosses zero n times and, therefore, has n roots.

Now, $P_n(x) - T_n(x)/2^{n-1}$ is a polynomial of degree $n - 1$, because $P_n(x)$ and $T_n(x)/2^{n-1}$ are monic polynomials.

Therefore, $P_n(x) - T_n(x)/2^{n-1}$ having n roots leads to a contradiction, implying that $P_n(x)$ can't exist.

Choosing the roots of $T_n(x)$ as the base points removes Runge phenomenon



$$x_i = \cos\left(\frac{(2i-1)\pi}{2n}\right), \text{ roots of } T_n(x)$$

For a general interval $[a, b]$

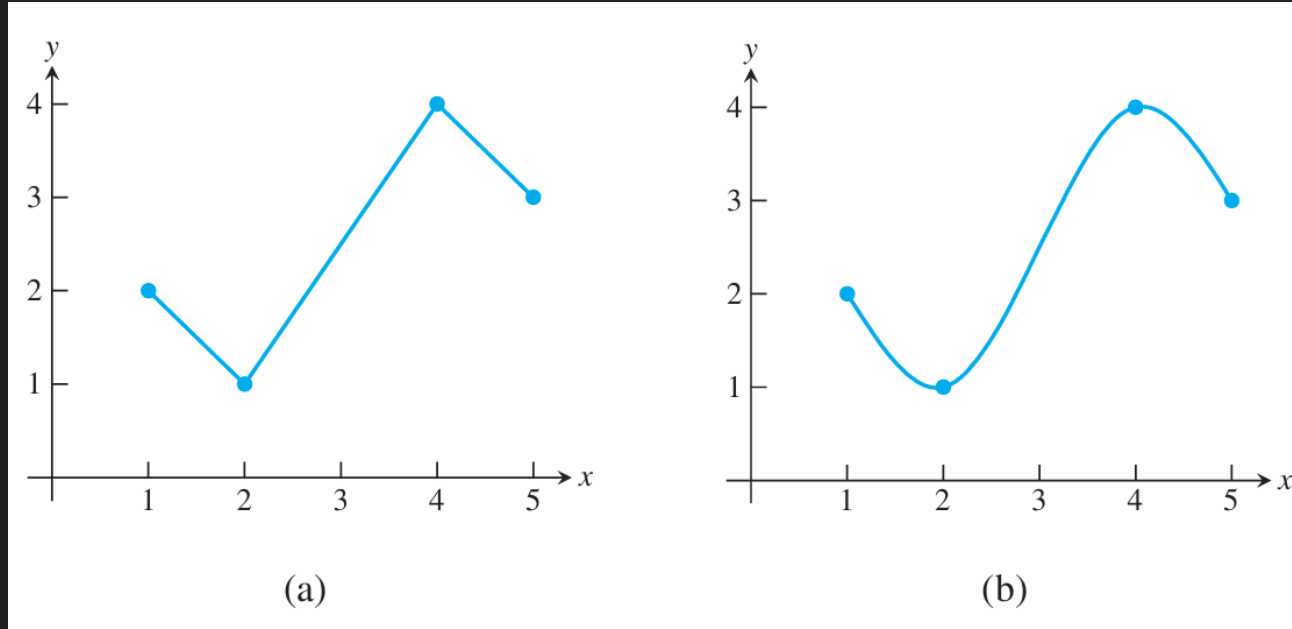
$$x_i = \frac{b-a}{2} \cos\left(\frac{(2i-1)\pi}{2n}\right) + \frac{b+a}{2}$$

Stretching

Translation



Cubic Splines (A spline is a function defined piecewise)



$$\begin{aligned} S_1(x) &= 2 - (x - 1) \text{ on } [1, 2] \\ S_2(x) &= 1 + \frac{3}{2}(x - 2) \text{ on } [2, 4] \\ S_3(x) &= 4 - (x - 4) \text{ on } [4, 5]. \end{aligned}$$

$$\begin{aligned} S_1(x) &= 2 - \frac{13}{8}(x - 1) + 0(x - 1)^2 + \frac{5}{8}(x - 1)^3 \text{ on } [1, 2] \\ S_2(x) &= 1 + \frac{1}{4}(x - 2) + \frac{15}{8}(x - 2)^2 - \frac{5}{8}(x - 2)^3 \text{ on } [2, 4] \\ S_3(x) &= 4 + \frac{1}{4}(x - 4) - \frac{15}{8}(x - 4)^2 + \frac{5}{8}(x - 4)^3 \text{ on } [4, 5]. \end{aligned}$$

Interpolation function $P(x) = S_1 \cup S_2 \cup S_3$

Properties of cubic splines:

Property 1 $S_i(x_i) = y_i$ and $S_i(x_{i+1}) = y_{i+1}$ for $i = 1, \dots, n - 1$.

Property 2 $S'_{i-1}(x_i) = S'_i(x_i)$ for $i = 2, \dots, n - 1$.

Property 3 $S''_{i-1}(x_i) = S''_i(x_i)$ for $i = 2, \dots, n - 1$.

Determining the constants in the splines:

$$S_1(x) = y_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 \text{ on } [x_1, x_2]$$

$$S_2(x) = y_2 + b_2(x - x_2) + c_2(x - x_2)^2 + d_2(x - x_2)^3 \text{ on } [x_2, x_3]$$

$\vdots$

$$S_{n-1}(x) = y_{n-1} + b_{n-1}(x - x_{n-1}) + c_{n-1}(x - x_{n-1})^2 + d_{n-1}(x - x_{n-1})^3 \text{ on } [x_{n-1}, x_n]$$

- $3(n-1)=3n-3$ unknown constants.
- Property 1 provides $n-1$ independent equations.
- Property 2 and 3 provide $n-2$ independent equations each.
- In total there are $2(n-2)+n-1=3n-5$ independent equations.
- $S_1''(x_1)=0$ and $S_{n-1}''(x_n)=0$ provide the other two necessary equations to uniquely determine the constants.

Define: $\delta_i = x_{i+1} - x_i$ and $\Delta_i = y_{i+1} - y_i$.

One can derive the following relations using the properties of splines

$$b_i = \frac{\Delta_i}{\delta_i} - c_i \delta_i - d_i \delta_i^2$$

$$d_i = \frac{c_{i+1} - c_i}{3\delta_i} \quad \text{for } i = 1, \dots, n-1.$$

Solving for c_i 's:

$$\begin{bmatrix} 1 & 0 & 0 & & \\ \delta_1 & 2\delta_1 + 2\delta_2 & \delta_2 & \ddots & \\ 0 & \delta_2 & 2\delta_2 + 2\delta_3 & \delta_3 & \\ & \ddots & \ddots & \ddots & \ddots \\ & & \delta_{n-2} & 2\delta_{n-2} + 2\delta_{n-1} & \delta_{n-1} \\ & & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 3\left(\frac{\Delta_2}{\delta_2} - \frac{\Delta_1}{\delta_1}\right) \\ \vdots \\ 3\left(\frac{\Delta_{n-1}}{\delta_{n-1}} - \frac{\Delta_{n-2}}{\delta_{n-2}}\right) \\ 0 \end{bmatrix}.$$

Check pages 169-171 in the reference book (by Timothy Sauer) for details.